Review

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Definition

- Given a real number a and a natural number n, we define a^n by
$$a^n = a \times \cdots \times a \quad (n \text{ factors})$$
- In the expression a^n the number a is the base, and n is the exponent or power to which the base is raised.

Properties of exponents

- $a^n a^m = a^{n+m}$
- $(a^n)^m = a^{mn}$
- $\frac{a^m}{a^n} = a^{m-n}$ if $m > n$
- $\frac{a^m}{a^n} = \frac{1}{a^{m-n}}$ if $m < n$
- $\frac{a^m}{a^n} = 1$ if $m = n$
- $(ab)^m = a^m b^m$
- $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

Properties of exponents - Example

- $a^5 a^6 = a^{11}$; $(x+1)(x+1)^2 = (x+1)^3$
- $(2^3)^4 = 2^{12}$; $[(x+1)^2[^3 = (x+1)^6$
- $\frac{a^6}{a^2} = a^4$; $\frac{a^2}{a^6} = \frac{1}{a^4}$; $\frac{a^5}{a^5} = 1$
- $(2x^2)^3 = 2^3 \times (x^2)^3 = 8x^6$; $\left(\frac{x^2}{y^3}\right)^4 = \frac{x^8}{y^{12}}$

Zero exponent and negative exponents

- For any nonzero real number a,

$$a^0 = 1$$

- 0^0 is undefined
- For any nonzero real number a and natural number n,

$$a^{-n} = \frac{1}{a^n}$$

Negative exponents - Examples

- $2^{-1} = \frac{1}{2}$
- $\left(\frac{1}{10}\right)^{-1} = \frac{1}{\frac{1}{10}} = 10$
- $x^{-2} = \frac{1}{x^2}$
- $(a^2b)^{-3} = \frac{1}{(a^2b)^3} = \frac{1}{a^6b^3}$

Example

Simplify the following expression. Write the answer in such a way that only positive exponents appear.

$$(a^2b^3)^2(a^5b)^{-1}$$

Solution

$(a^2b^3)^2(a^5b)^{-1} = (a^2b^3)^2\frac{1}{a^5b} = \frac{a^4b^6}{a^5b} = \frac{b^{6-1}}{a^{5-4}} = \frac{b^5}{a}$

Alternatively

$(a^2b^3)^2(a^5b)^{-1} = (a^4b^6)(a^{-5}b^{-1}) = a^{4+-5}b^{6+-1} = a^{-1}b^5 = \frac{1}{a} \times b^5 = \frac{b^5}{a}$

Example

Simplify the following expression, writing the answer so that negative exponents are not used.

$$\left(\frac{a^{-5}b^2c^0}{a^3b^{-1}}\right)^3$$

Solution

$$\left(\frac{a^{-5}b^2c^0}{a^3b^{-1}}\right)^3 = \frac{a^{-15}b^6}{a^9b^{-3}} = \frac{b^{6-(-3)}}{a^{9-(-15)}} = \frac{b^9}{a^{24}}$$

Alternatively

$$\left(\frac{a^{-5}b^2c^0}{a^3b^{-1}}\right)^3 = \left(\frac{b^{2-(-1)}}{a^{3-(-5)}}\right)^3 = \left(\frac{b^3}{a^8}\right)^3 = \frac{b^9}{a^{24}}$$

Example

Simplify the following expressions, writing the answers so that negative exponents are not used. (Assume that p and q are natural numbers.)

- $\frac{a^{4p+q}}{a^{p-q}}$
- $(a^p b^q)^2 (a^{3p} b^{-q})^{-1}$

Solution

- $\frac{a^{4p+q}}{a^{p-q}} = a^{(4p+q)-(p-q)} = a^{3p+2q}$
- $(a^p b^q)^2 (a^{3p} b^{-q})^{-1} = (a^{2p} b^{2q})(a^{-3p} b^q) = a^{2p-3p} b^{2q+q} = a^{-p} b^{3q} = \frac{b^{3q}}{a^p}$

Example

Use the properties of exponents to compute the quantity $\frac{2^{10}\times 3^{13}}{27\times 6^{12}}$

Solution

$\frac{2^{10}\times 3^{13}}{27\times 6^{12}} = \frac{2^{10}\times 3^{13}}{3^3\times (2\times 3)^{12}} = \frac{2^{10}\times 3^{13}}{3^3\times 2^{12}\times 3^{12}} = \frac{2^{10}\times 3^{13}}{2^{12}\times 3^{15}} = \frac{1}{2^{(12-10)}3^{(15-13)}} = \frac{1}{2^2\times 3^2} = \frac{1}{36}$

Scientific Notation

- To write any positive number in scientific notation, we express it as a number between 1 and 10, multiplied by an appropriate power of 10. That is, we write it in the form

$$a \times 10^n$$

where $1 \leq a < 10$ and n is an integer

- To convert a given number into scientific notation, well rely on the following two-step procedure
  - First move the decimal point until it is to the immediate right of the first nonzero digit
  - Then multiply by 10^n or 10^{-n}, depending on whether the decimal point was moved n places to the left or to the right, respectively

Scientific Notation

- For example, to express the number 29,979,000,000 in scientific notation, first we move the decimal point 10 places to the left so that its located between the 2 and the 9, then we multiply by 10^{10}. The result is

$$29,979,000,000 = 2.9979 \times 10^{10}$$

nth Roots

nth Roots - Definition

- Let n be a natural number. If a and b are real numbers and

$$a^n = b$$

  then we say that a is an nth root of b.
- When $n = 2$ and when $n = 3$, we refer to the roots as square roots and cube roots, respectively

nth Roots - Examples

- Both 3 and -3 are square roots of 9 because $3^2 = 9$ and $(-3)^2 = 9$.
- Both 2 and -2 are fourth roots of 16 because $2^4 = 16$ and $(-2)^4 = 16$.
- -3 is a fifth root of -243 because $(-3)^5 = -243$.

nth Roots

- We use the notation $\sqrt[n]{b}$ to denote the positive, nth root of b.
- The symbol $\sqrt{\ }$ is called a radical sign, and the number within the radical sign is the radicand.
- The natural number n used in the notation $\sqrt[n]{\ }$ is called the index of the radical
- For square roots, we suppress the index and simply write $\sqrt{\ }$ rather than $\sqrt[2]{\ }$

Properties of nth Roots

Suppose that x and y are real numbers and that m and n are natural numbers. Then each of the following properties holds, provided only that the expressions on both sides of the equation are defined (and so represent real numbers).

- $\left(\sqrt[n]{x}\right)^n = x$
- $\sqrt[n]{xy} = \sqrt[n]{x}\sqrt[n]{y}$
- $\sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$
- if n is even, $\sqrt[n]{x^n} = |x|$
- if n is odd, $\sqrt[n]{x^n} = x$
- $\sqrt[m]{\sqrt[n]{x}} = \sqrt[mn]{x}$

Simplifying Square Roots - Example

Simplify

- $\sqrt{12} + \sqrt{75}$
- $\sqrt{\frac{162}{49}}$

Simplifying Square Roots - Solution

- $\sqrt{12} + \sqrt{75} = \sqrt{4 \times 3} + \sqrt{25 \times 3} = 2\sqrt{3} + 5\sqrt{3} = 7\sqrt{3}$
- $\sqrt{\frac{162}{49}} = \sqrt{\frac{81 \times 2}{49}} = \frac{\sqrt{81 \times 2}}{\sqrt{49}} = \frac{9 \times \sqrt{2}}{7}$

Simplifying nth Roots - Example

Simplify $\sqrt[3]{16} + \sqrt[3]{250} - \sqrt[3]{128}$

Simplifying nth Roots - Solution

$\sqrt[3]{16} + \sqrt[3]{250} - \sqrt[3]{128} = \sqrt[3]{8 \times 2} + \sqrt[3]{125 \times 2} - \sqrt[3]{64 \times 2} =$
$2\sqrt[3]{2} + 5\sqrt[3]{2} - 4\sqrt[3]{2} = 3\sqrt[3]{2}$

Simplifying nth Roots - Example

Simplify each of the following expressions by removing the largest possible perfect-square or perfect-cube factor from within the radical:

- $\sqrt{8x^2}$
- $\sqrt{8x^2}$ where $x \geq 0$
- $\sqrt{18a^7}$ where $a \geq 0$
- $\sqrt[3]{16y^5}$

Simplifying nth Roots - Solution

- $\sqrt{8x^2} = \sqrt{4 \times 2 \times x^2} = 2|x|\sqrt{2}$
- $\sqrt{8x^2} = 2x\sqrt{2}$ where $x \geq 0$
- $\sqrt{18a^7} = \sqrt{9 \times 2 \times a^6 \times a} = \sqrt{9 \times 2 \times (a^3)^2 \times a} = 3a^3\sqrt{2a}$ where $a \geq 0$
- $\sqrt[3]{16y^5} = \sqrt[3]{8 \times 2 \times y^3 \times y^2} = 2y\sqrt[3]{2y^2}$

Simplifying nth Roots - Example

Simplify each of the following, assuming that a, b, and c are positive: :

- $\sqrt{8ab^2c^5}$
- $\sqrt[4]{\frac{32a^6b^5}{c^8}}$

Simplifying nth Roots - Solution

- $\sqrt{8ab^2c^5} = \sqrt{4 \times 2 \times a \times b^2 \times (c^2)^2 \times c} = 2bc^2\sqrt{2ac}$
- $\sqrt[4]{\frac{32a^6b^5}{c^8}} = \frac{\sqrt[4]{2^4 \times 2 \times a^4 \times a^2 \times b^4 \times b}}{\sqrt[4]{(c^2)^4}} = \frac{2ab\sqrt[4]{2a^2b}}{c^2}$

Rationalising numerator / denominator involving square roots

- To rationalize the denominator of , say $\sqrt{y}$, multiply by $\sqrt{y}$
- To rationalize an expression of the the form $a + \sqrt{b}$, multiply by $a - \sqrt{b}$ (Note that $a - \sqrt{b}$ is called the conjugate of $a + \sqrt{b}$). This follows from the observation that $x^2 - y^2 = (x - y)(x + y)$, as such $(a - \sqrt{b})(a + \sqrt{b}) = a^2 - (\sqrt{b})^2) = a^2 - b$

Rationalizing a square root denominator - Example

Simplify $\frac{1}{\sqrt{2}} - 3\sqrt{50}$

Rationalizing a square root denominator - Solution

We first rationalize the denominator of $\frac{1}{\sqrt{2}}$. This yields

$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$.

Next observe that $3\sqrt{50} = 3\sqrt{25 \times 2} = 15\sqrt{2}$.

Hence $\frac{1}{\sqrt{2}} - 3\sqrt{50} = \frac{\sqrt{2}}{2} - 15\sqrt{2} = \frac{\sqrt{2} - 30\sqrt{2}}{2} = \frac{-29\sqrt{\sqrt{2}}}{2}$

Using the conjugate to rationalize a denominator - Example

Simplify $\frac{4}{2+\sqrt{3}}$

Rationalizing a square root denominator - Solution

$$\frac{4}{2+\sqrt{3}} = \frac{4}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{4(2-\sqrt{3})}{4-3} = 8-4\sqrt{3}$$

Rationalizing a numerator using the conjugate - Example

Rationalize the numerator of $\frac{\sqrt{x}-\sqrt{3}}{x-3}$

Rationalizing a numerator using the conjugate - Solution

$$\frac{\sqrt{x}-\sqrt{3}}{x-3} = \frac{\sqrt{x}-\sqrt{3}}{x-3} \times \frac{\sqrt{x}+\sqrt{3}}{\sqrt{x}+\sqrt{3}} = \frac{x-3}{(x-3)(\sqrt{x}+\sqrt{3})} = \frac{1}{\sqrt{x}+\sqrt{3}}$$

Rationalizing denominators containing nth roots

To rationalize a numerator or a denominator involving an nth root, we multiply the numerator or denominator by a factor that yields a product that itself is a perfect nth power

Rationalizing denominators containing nth roots - Example

Rationalize the denominator of

- $\frac{6}{\sqrt[3]{7}}$
- $\frac{ab}{\sqrt[4]{a^2b^3}}$

Rationalizing denominators containing nth roots - Solution

- $\frac{6}{\sqrt[3]{7}} = \frac{6}{\sqrt[3]{7}} \times \frac{\sqrt[3]{7^2}}{\sqrt[3]{7^2}} = \frac{6\sqrt[3]{7^2}}{\sqrt[3]{7^3}} = \frac{6\sqrt[3]{7^2}}{7}$
- $\frac{ab}{\sqrt[4]{a^2b^3}} = \frac{ab}{\sqrt[4]{a^2b^3}} \times \frac{\sqrt[4]{a^2b}}{\sqrt[4]{a^2b}} = \frac{ab\sqrt[4]{a^2b}}{\sqrt[4]{a^4b^4}} = \frac{ab\sqrt[4]{a^2b}}{ab} = \sqrt[4]{a^2b}$

Rationalizing denominators containing nth roots

The following becomes helpful when rationalizing expressions involving sums or differences of cubes;
$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$ and
$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$

Using the conjugate to rationalize a denominator involving a cube root - Example

Rationalize the denominator in the expression $\frac{1}{\sqrt[3]{a}+\sqrt[3]{b}}$

Using the conjugate to rationalize a denominator involving a cube root - Solution

Observe that the conjugate of $\sqrt[3]{a}+\sqrt[3]{b}$ is $(\sqrt[3]{a})^2-\sqrt[3]{a}\sqrt[3]{b}+(\sqrt[3]{b})^2=\sqrt[3]{a^2}-\sqrt[3]{ab}+\sqrt[3]{b^2}$.

Therefore $\frac{1}{\sqrt[3]{a}+\sqrt[3]{b}}=\frac{1}{\sqrt[3]{a}+\sqrt[3]{b}}\times\frac{\sqrt[3]{a^2}-\sqrt[3]{ab}+\sqrt[3]{b^2}}{\sqrt[3]{a^2})-\sqrt[3]{ab}+\sqrt[3]{b^2}}=\frac{\sqrt[3]{a^2}-\sqrt[3]{ab}+\sqrt[3]{b^2}}{(\sqrt[3]{a})^3+\sqrt[3]{b})^3}=\frac{\sqrt[3]{a^2}-\sqrt[3]{ab}+\sqrt[3]{b^2}}{a+b}$

Rational Exponents

Rational Exponents - Definition

Let b denote a real number and n a natural number. We define $b^{\frac{1}{n}}$ by

$$b^{\frac{1}{n}} = \sqrt[n]{b}$$

By extension

$$b^{\frac{m}{n}} = (b^{\frac{1}{n}})^m = (\sqrt[n]{b})^m = \sqrt[n]{b^m}$$

where m and n are natural numbers

Evaluating rational exponents - Example

Simplify each of the following quantities. Express the answers using positive exponents. If an expression does not represent a real number, say so.

- $49^{\frac{1}{2}}$
- $-49^{\frac{1}{2}}$
- $(-49)^{\frac{1}{2}}$
- $49^{-\frac{1}{2}}$

Evaluating rational exponents - Solution

- $49^{\frac{1}{2}} = \sqrt{49} = 7$
- $-49^{\frac{1}{2}} = -(49^{\frac{1}{2}}) = -(\sqrt{49}) = -7$
- $(-49)^{\frac{1}{2}} = \sqrt{-49}$ which does not represent a real number
- $49^{-\frac{1}{2}} = \frac{1}{49^{\frac{1}{2}}} = \frac{1}{\sqrt{49}} = \frac{1}{7}$

Simplifying expressions containing rational exponents - Example

Simplify each of the following. Write the answers using positive exponents. (Assume that $a > 0$.)

- $(5a^{\frac{2}{3}})(4a^{\frac{3}{4}})$
- $\sqrt[5]{\frac{16a^{\frac{1}{3}}}{a^{\frac{1}{4}}}}$
- $(x^2+1)^{\frac{1}{5}}(x^2+1)^{\frac{4}{5}}$

Simplifying expressions containing rational exponents - Solution

- $(5a^{\frac{2}{3}})(4a^{\frac{3}{4}}) = 20a^{\frac{2}{3}+\frac{3}{4}} = 20a^{\frac{17}{12}}$
- $\sqrt[5]{\frac{16a^{\frac{1}{3}}}{a^{\frac{1}{4}}}} = \left(\frac{16a^{\frac{1}{3}}}{a^{\frac{1}{4}}}\right)^{\frac{1}{5}} = (16a^{\frac{1}{3}-\frac{1}{4}})^{\frac{1}{5}} = (16a^{\frac{1}{12}})^{\frac{1}{5}} = 16^{\frac{1}{5}}a^{\frac{1}{60}}$
- $(x^2+1)^{\frac{1}{5}}(x^2+1)^{\frac{4}{5}} = (x^2+1)^{\frac{1}{5}+\frac{4}{5}} = (x^2+1)^1 = x^2+1$

Evaluating rational exponents - Example

Simplify:

- $32^{-\frac{2}{5}}$
- $(-8)^{\frac{4}{3}}$

Evaluating rational exponents - Solution

- $32^{-\frac{2}{5}} = (32^{\frac{1}{5}})^{-2} = (\sqrt[5]{32})^{-2} = 2^{-2} = \frac{1}{2^2} = \frac{1}{4}$
  Altenatively $32^{-\frac{2}{5}} = (2^5)^{-\frac{2}{5}} = 2^{-2} = \frac{1}{2^2} = \frac{1}{4}$
- $(-8)^{\frac{4}{3}} = ((-8)^{\frac{1}{3}})^4 = (\sqrt[3]{-8})^4 = (-2)^4 = 16$
  Alternatively $(-8)^{\frac{4}{3}} = ((-2)^3)^{\frac{4}{3}} = (-2)^4 = 16$

Using rational exponents to combine radicals - Example

Consider the expression $\sqrt{x}\sqrt[3]{y^2}$ where x and y are positive

- Rewrite the expression using rational exponents
- Rewrite the expression using only one radical sign.

Using rational exponents to combine radicals - Solution

- $\sqrt{x}\sqrt[3]{y^2} = x^{\frac{1}{2}}y^{\frac{2}{3}}$
- $\sqrt{x}\sqrt[3]{y^2} = x^{\frac{1}{2}}y^{\frac{2}{3}} = x^{\frac{3}{6}}y^{\frac{4}{6}} = (x^3y^4)^{\frac{1}{6}} = \sqrt[6]{x^3y^4}$

Factoring Polynomials

Factoring Polynomials

- To factor a polynomial means to write it as a product of two or more nonconstant polynomials, for instance the factorization of $x^2 - 9$ is given by $x^2 - 9 = (x - 3)(x + 3)$
- For now, if the polynomial or expression that we wish to factor contains only integer coefficients, then the factors (if any) should involve only integer coefficients. Thus we will not consider such factorizations as $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$ since it involves coefficients that are irrational numbers....PLEASE NOTE THAT THIS IS JUST FOR NOW
- As it happens, $x^2 - 2$ is an example of a polynomial that cannot be factored using integer coefficients. In such a case we say that the polynomial is **irreducible** over the integers.
- If the coefficients of a polynomial are rational numbers, then we will allow factors with rational coefficients

Basic Factoring Techniques

- Look for common factors (with the highest degree). For instance $3x^4 + 6x^3 - 12x^2 = 3x^2(x^2 + 2x - 4)$ and $4(x^2 + 1) - x(x^2 + 1) = (x^2 + 1)(4 - x)$
- Difference of squares. i.e $x^2 - a^2 = (x - a)(x + a)$
- Trial and error
- Difference of cubes. i.e. $x^3 - a^3 = (x - a)(x^2 + ax + a^2)$
- Sum of cubes, i.e. $x^3 + a^3 = (x + a)(x^2 - ax + a^2)$
- Grouping, for instance $x^3 - x^2 + x - 1 = (x^3 - x^2) + (x - 1) = x^2(x - 1) + 1(x - 1) = (x - 1)(x^2 + 1)$

Factoring using the difference-of-squares technique - Example

Factor

- $x^2 - 49$
- $(2a - 3b)^2 - 49$

Factoring using the difference-of-squares technique - Solution

- $x^2 - 49 = x^2 - 7^2 = (x - 7)(x + 7)$
- $(2a-3b)^2-49 = (2a-3b)^2-7^2 = ((2a-3b)-7)((2a-3b)+7)$

Factoring using common factor and difference of squares - Example

Factor

- $2x^3 - 50x$
- $3x^5 - 3x$

Factoring using common factor and difference of squares - Solution

- $2x^3 - 50x = 2x(x^2 - 25) = 2x(x^2 - 5^2) = 2x(x - 5)(x + 5)$
- $3x^5 - 3x = 3x(x^4 - 1) = 3x((x^2)^2 - 1^2) =$
  $3x(x^2 - 1)(x^2 + 1) = 3x(x - 1)(x + 1)(x^2 + 1)$

Factoring by trial and error - Example

Factor

- $x^2 - 4x - 5$

Factoring by trial and error - Solution

- We expect the factorization to take the form $(x + ...)(x + ...)$ where slot denoted ... will factors of -5. So we have the posibilities;
  $(x + (-5))(x + 1) = (x - 5)(x + 1)$ and
  $(x + (-1))(x + 5) = (x - 1)(x + 5)$
  Multipying out we observe that the following is the correct factorization
  $x^2 - 4x - 5 = (x - 5)(x + 1)$

Factoring by trial and error - Example

Factor

- $(a+b)^2 - 4(a+b) - 5$

Factoring by trial and error - Solution

- Let $a+b=y$, then
  $(a+b)^2 - 4(a+b) - 5 = y^2 - 4y - 5 = (y-5)(y+1)$.
  Now since $a+b=y$, back-substitution yields
  $(a+b)^2 - 4(a+b) - 5 = ((a+b)-5)((a+b)+1) =$
  $(a+b-5)(a+b+1)$

Factoring by trial and error - Example

Factor $2z^4 + 9z^2 + 4$

Factoring by trial and error - Solution

The posibilities are $(2z^2 + 2)(z^2 + 2)$, $(2z^2 + 1)(z^2 + 4)$, $(2z^2 + 4)(z^2 + 1)$. Mutliplying out each one, we observe that the correct one is $(2z^2 + 1)(z^2 + 4)$, hence
$2z^4 + 9z^2 + 4 = (2z^2 + 1)(z^2 + 4)$

Factoring using grouping - Example

Factor $x^2 - y^2 + 10x + 25$

Factoring using grouping - Solution

We observe that $x^2 + 10x + 25 = (x + 5)^2$, therefore we group the terms of the expression as follows

$x^2 - y^2 + 10x + 25 = (x^2 + 10x + 25) - y^2 = (x + 5)^2 - y^2 = ((x + 5) - y)((x + 5) + y) = (x - y + 5)(x + y + 5)$

Factoring using grouping - Example

Factor $ax + ay^2 + bx + by^2$

Factoring using grouping - Solution

$ax + ay^2 + bx + by^2 = (ax + ay^2) + (bx + by^2) =$
$a(x + y^2) + b(x + y^2) = (a + b)(x + y^2)$
Alternatively
$ax + ay^2 + bx + by^2 = (ax + bx) + (ay^2 + by^2) =$
$x(a + b) + y^2(a + b) = (a + b)(x + y^2)$

Factoring sum and difference of cubes - Example

Factor

- $t^3 - 125$
- $8 + (a-2)^3$

Factoring sum and difference of cubes - Solution

- $t^3 - 125 = t^3 - 5^3 = (t-5)(t^2 + 5t + 25)$
- Let $y = a - 2$, then
  $8 + (a-2)^3 = 8 + y^3 = 2^3 + y^3 = (2+y)(2^2 - 2y + y^2)$.
  Now since $y = a - 2$, back-substitution yields
  $8 + (a-2)^3 = 2^3 + (a-2)^3 = (2 + (a-2))(2^2 - 2(a-2) + (a-2)^2) = a(4 - 2a + 4 + a^2 - 4a + 4) = a(a^2 - 6a + 12)$

Factoring with rational exponents - Example

Factor $x(2x-1)^{-\frac{1}{2}} + (2x-1)^{\frac{3}{2}}$

Factoring with rational exponents - Solution

Let $y = (2x-1)$, so $x(2x-1)^{-\frac{1}{2}} + (2x-1)^{\frac{3}{2}} = xy^{-\frac{1}{2}} + y^{\frac{3}{2}} = y^{-\frac{1}{2}}\left(x + \frac{y^{\frac{3}{2}}}{y^{-\frac{1}{2}}}\right) = y^{-\frac{1}{2}}\left(x + y^{\frac{3}{2}-(-\frac{1}{2})}\right) = y^{-\frac{1}{2}}(x+y^2)$

Back-substitution yields

$x(2x-1)^{-\frac{1}{2}} + (2x-1)^{\frac{3}{2}} = (2x-1)^{-\frac{1}{2}}(x + (2x-1)^{\frac{3}{2}-(-\frac{1}{2})}) = (2x-1)^{-\frac{1}{2}}(x + (2x-1)^{\frac{3}{2}+\frac{1}{2}}) = (2x-1)^{-\frac{1}{2}}(x + (2x-1)^2) = (2x-1)^{-\frac{1}{2}}(4x^2 - 3x + 1)$

Note: The trick is to factor out the factor with the smaller exponent

Fractional Expressions

Negatives and fractions - Properties

- $b - a = -(a - b)$
- $\frac{b-a}{a-b} = \frac{b-a}{-(b-a)} = -1$

Negatives and fractions - Example

$$\frac{4-3x}{15x-20} = \frac{4-3x}{5(3x-4)} = \frac{4-3x}{-5(4-3x)} = -\frac{1}{5}$$

Using factoring to simplify a fractional expression - Example

Simplify

- $\frac{x^3-8}{x^2-2x}$
- $\frac{x^2-6x+8}{a(x-2)+b(x-2)}$

Using factoring to simplify a fractional expression - Solution

- $\frac{x^3-8}{x^2-2x} = \frac{(x-2)(x+2x+4)}{x(x-2)} = \frac{x+2x+4}{x}$
- $\frac{x^2-6x+8}{a(x-2)+b(x-2)} = \frac{(x-4)(x-2)}{(x-2)(a+b)} = \frac{x-4}{a+b}$

Using factoring to simplify products and quotients - Example

Simplify

- $\frac{x^3}{2x^2+3x} \times \frac{12x+18}{4x^2-6x}$
- $\frac{x^2-144}{x^2-4} \div \frac{x+12}{x+2}$

Using factoring to simplify products and quotients - Solution

- $\frac{x^3}{2x^2+3x} \times \frac{12x+18}{4x^2-6x} = \frac{x^3}{x(2x+3)} \times \frac{6(2x+3)}{2x(2x-3)} = \frac{3x}{2x-3}$
- $\frac{x^2-144}{x^2-4} \div \frac{x+12}{x+2} = \frac{x^2-144}{x^2-4} \times \frac{x+2}{x+12} = \frac{(x-12)(x+12)}{(x-2)(x+2)} \times \frac{x+2}{x+12} = \frac{x-12}{x-2}$

Adding and subtracting Fractional Expressions

- To combine two fractions by addition or subtraction, the fractions must have a common denominator, that is, the denominators must be the same
- In that case then we have $\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$ and $\frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}$
- Fractions with unlike denominators are added or subtracted by first converting to a common denominator. Thus to add $\frac{x}{a}$ and $\frac{y}{a^2}$, convert $\frac{x}{a}$ to have denominator a^2. To do that, multiply the numerator and denominator by a to get $\frac{x}{a} \times \frac{a}{a} = \frac{ax}{a^2}$. Hence $\frac{x}{a} + \frac{y}{a^2} = \frac{ax}{a^2} + \frac{y}{a^2} = \frac{ax+y}{a^2}$. In this case a^2 is called the least common denominator

Using the least common denominator - Example

Combine into a single fraction and simplify

- $\frac{3}{4x} + \frac{7x}{10y^2} - 2$
- $\frac{x}{x^2-9} - \frac{1}{x+3}$

Using the least common denominator - Solution

- The denominators are $2^2x, 2 \times 5 \times y^2, 1$. Hence the least common denominator is $2^2x \times 5 \times y^2 = 20xy^2$. Hence $\frac{3}{4x} + \frac{7x}{10y^2} - 2 = \frac{3}{4x} \times \frac{5y^2}{5y^2} + \frac{7x}{10y^2} \times \frac{2x}{2x} - \frac{2}{1} \times \frac{20xy^2}{20xy^2} = \frac{15y^2}{20xy^2} + \frac{14x^2}{20xy^2} - \frac{40xy^2}{20xy^2} = \frac{15y^2+14x^2-40xy^2}{20xy^2}$
- The denominators are $(x^2 - 9) = (x - 3)(x + 3), (x + 3)$, so the least common denominator is $(x^2 - 9) = (x - 3)(x + 3)$. Hence $\frac{x}{x^2-9} - \frac{1}{x+3} = \frac{x}{x^2-9} - \frac{1}{x+3} \times \frac{x-3}{x-3} = \frac{x}{x^2-9} - \frac{x-3}{x^2-9} = \frac{x-(x-3)}{x^2-9} = \frac{3}{x^2-9}$

Simplifying a compound fraction - Example

Simplify $\frac{\frac{1}{3a}+\frac{1}{4b}}{\frac{5}{6a^2}+\frac{1}{b}}$

Simplifying a compound fraction - Solution

The denominators are $3 \times a, 2^2 \times b, 2 \times 3 \times a^2, b$, hence the least common denominator is $2^2 \times 2 \times a^2 \times b = 12a^2b$. Now for such a compound fraction we multiply both the numerator and denominator by $12a^2b$. Hence

$$\frac{\frac{1}{3a}+\frac{1}{4b}}{\frac{5}{6a^2}+\frac{1}{b}} = \frac{12a^2b}{12a^2b} \times \frac{\frac{1}{3a}+\frac{1}{4b}}{\frac{5}{6a^2}+\frac{1}{b}} = \frac{4ab+3a^2}{10b+12a^2}$$

Simplifying a fraction containing negative exponents - Example

Simplify $(x^{-1} + y^{-1})^{-1}$

Simplifying a fraction containing negative exponents - Solution

$(x^{-1} + y^{-1})^{-1} = \frac{1}{x^{-1}+y^{-1}} = \frac{1}{\frac{1}{x}+\frac{1}{y}}$.

Clearly the least common denominator is xy, so multiplying the denominator and numerator by the least common denominator yields

$(x^{-1} + y^{-1})^{-1} = \frac{1}{x^{-1}+y^{-1}} = \frac{1}{\frac{1}{x}+\frac{1}{y}} = \frac{xy}{xy} \times \frac{1}{\frac{1}{x}+\frac{1}{y}} = \frac{xy}{y+x}$.

Simplifying a compound fraction - Example

Simplify $\frac{x-\frac{1}{x^2}}{\frac{1}{x^2}-1}$

Simplifying a compound fraction - Solution

The least common denominator is x^2, so

$$\frac{x-\frac{1}{x^2}}{\frac{1}{x^2}-1} = \frac{x^2}{x^2} \times \frac{x-\frac{1}{x^2}}{\frac{1}{x^2}-1} = \frac{x^3-1}{1-x^2} = \frac{(x-1)(x^2+x+1^2)}{(1-x)(1+x)} = -\frac{x^2+x+1}{1+x}$$